

An Abstract Isometric Isomorphism Need Not Imply Exterior Equivalence of Twisted L^p -Actions

Counterexample packet for arXiv:2509.24106

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Status: counterexample likely valid. The literal abstract-isomorphism converse is false. The construction does not address a strengthened converse requiring the isomorphism to fix the coefficient algebra or preserve a grading/coaction.

1 Source question

Delfin, Farsi, and Packer define reduced twisted L^p crossed products and ask the following in the introduction of [1], on PDF page 4:

For $p \neq 2$, does an isometric isomorphism between $F_r^p(G, A, \alpha, \sigma)$ and $F_r^p(G, A, \beta, \omega)$ imply that (α, σ) and (β, ω) are exterior equivalent?

Finally, we also mention that Theorem [6.3](#) also holds for the reduced twisted crossed product $F_r^p(G, A, \alpha, \sigma)$. An interesting question, which we do not tackle in this project, is whether the converse holds when $p \neq 2$. That is, does an isometric isomorphism between $F_r^p(G, A, \alpha, \sigma)$ and $F_r^p(G, A, \beta, \omega)$ imply that (α, σ) and (β, ω) are exterior equivalent? A positive answer to this problem will add to the recent list of rigidity results for $p \neq 2$ such as [\[19\]](#) Proposition 5.5], [\[6\]](#) Theorem 6.7], and [\[20\]](#) Theorem 4.9].

Figure 1: The question in the original paper, PDF page 4.

The wording imposes no requirement that the isomorphism fix the canonical copy of A or preserve additional crossed-product structure. We answer this literal question.

2 Definitions and construction

Fix any $p \in [1, \infty) \setminus \{2\}$. Let

$$G = C_2 = \{e, s\}, \quad A = C(\{1, 2, 3\}) \cong \mathbb{C}^3$$

with the supremum norm. Represent A diagonally and nondegenerately on ℓ_3^p . Thus A is a unital, nondegenerate, separably representable L^p -operator algebra.

For a permutation u of $\{1, 2, 3\}$, write

$$\gamma^u(a) = a \circ u^{-1} \quad (a \in A).$$

Then $\gamma^u \gamma^v = \gamma^{uv}$. Put

$$r = (12), \quad q = (23), \quad t = (123),$$

and define two untwisted actions by

$$\alpha_e = \beta_e = \text{id}_A, \quad \alpha_s = \gamma^r, \quad \beta_s = \gamma^q, \quad \sigma = \omega = \mathbf{1}.$$

Since $q = trt^{-1}$, the isometric coefficient automorphism $\phi = \gamma^t$ conjugates the actions:

$$\beta_x = \phi \alpha_x \phi^{-1} \quad (x \in G). \quad (1)$$

For a discrete group with trivial twist, the algebraic convolution is

$$(f *_{\alpha} h)(x) = \sum_{y \in G} f(y) \alpha_y(h(y^{-1}x)).$$

Define

$$(\Phi f)(x) = \phi(f(x)), \quad f \in C_c(G, A). \quad (2)$$

3 Proof intuition

The source implication cannot distinguish exterior equivalence from ordinary conjugacy of the action unless the crossed-product isomorphism is required to remember the coefficient coordinates. Conjugating an action by an isometric automorphism ϕ of A makes $f \mapsto \phi \circ f$ an algebraic crossed-product isomorphism. For the reduced norm, precomposition with ϕ^{-1} pairs every coefficient representation for the first action with one for the second action; the paired regular integrated operators are literally equal.

The finite commutative algebra $A = \mathbb{C}^3$ makes the obstruction decisive. Two different coordinate actions can be conjugate, but exterior equivalence can only modify an action by inner multiplier automorphisms. Every such inner automorphism is trivial on a commutative unital algebra.

4 Counterexample theorem

Theorem 1. *For every $p \in [1, \infty) \setminus \{2\}$ there are a second countable locally compact group G , a nondegenerate separably representable L^p -operator algebra A with a contractive approximate identity, and two twisted actions $(\alpha, \mathbf{1})$ and $(\beta, \mathbf{1})$ on A such that*

$$F_r^p(G, A, \alpha, \mathbf{1}) \cong F_r^p(G, A, \beta, \mathbf{1})$$

isometrically as Banach algebras, while the two twisted actions are not exterior equivalent.

Proof. Use the construction above. First, (1) gives, for $f, h \in C_c(G, A)$,

$$\begin{aligned} \Phi(f *_{\alpha} h)(x) &= \sum_{y \in G} \phi(f(y)) \phi \alpha_y(h(y^{-1}x)) \\ &= \sum_{y \in G} \Phi f(y) \beta_y(\Phi h(y^{-1}x)) = (\Phi f *_{\beta} \Phi h)(x). \end{aligned}$$

Thus Φ is an algebra isomorphism with inverse obtained from ϕ^{-1} . It is also an isometry for the $L^1(G; A)$ norm.

We now verify the reduced norm. In the notation of [1], let $\pi_0 : A \rightarrow \mathcal{B}(E)$ be any nondegenerate, σ -finite, contractive representation on an L^p -space, and define

$$\rho_0 = \pi_0 \circ \phi^{-1}.$$

This operation is a bijection on the class of coefficient representations used to define the reduced crossed-product norms. Let $\Pi_{\pi_0}^\alpha$ and $\Pi_{\rho_0}^\beta$ denote the corresponding regular integrated representations on $L^p(G; E)$. The regular coefficient operator for the first action is given pointwise by

$$(\pi_{\pi_0}^\alpha(a)\xi)(x) = \pi_0(\alpha_x^{-1}(a))\xi(x).$$

Using (1), the paired coefficient operator satisfies

$$\begin{aligned} \rho_0(\beta_x^{-1}(\phi(a))) &= \pi_0(\phi^{-1}\beta_x^{-1}\phi(a)) \\ &= \pi_0(\alpha_x^{-1}(a)). \end{aligned}$$

Both twists are trivial, so in both regular representations the group operator for $y \in G$ is the same left translation $\xi(x) \mapsto \xi(y^{-1}x)$. It follows that

$$\Pi_{\rho_0}^\beta(\Phi f) = \Pi_{\pi_0}^\alpha(f) \quad (f \in C_c(G, A)). \quad (3)$$

Taking operator norms and then the supremum over the paired coefficient representations gives

$$\|\Phi f\|_{r,\beta} = \|f\|_{r,\alpha}.$$

Consequently Φ extends to the claimed isometric algebra isomorphism of reduced crossed products.

It remains to disprove exterior equivalence. The algebra A is unital and commutative, so $M(A) = A$ is commutative. Hence $\text{Ad}(\theta_x) = \text{id}_A$ for every invertible multiplier θ_x . The first algebraic requirement for an exterior equivalence in [1] would therefore give $\beta_x = \alpha_x$ for all x . This is false: for the first coordinate idempotent $e_1 = (1, 0, 0)$, $\alpha_s(e_1) = e_2$ whereas $\beta_s(e_1) = e_1$. Thus the twisted actions are not exterior equivalent. $\square$

5 Verification notes

The script `code/verify_finite_model.py` checks exactly that $q = trt^{-1}$, that the actions differ, and that Φ preserves all 36 products of the standard basis of the six-dimensional algebra $C_c(C_2, \mathbb{C}^3)$. It also checks (3) on every algebraic basis element for the faithful diagonal coefficient representation. The general reduced-norm conclusion is not inferred from this finite test; it follows from the representation-class bijection in the proof.

No amenability theorem or full-versus-reduced identification is used. The group C_2 and the algebra $\mathbb{C}^3$ satisfy all standing hypotheses directly.

6 Novelty check, limitations, and review recommendation

A bounded search on 22 July 2026 covered the run's cheap result indexes, the exact arXiv id and title, and arXiv queries combining the phrases *twisted L^p crossed product*, *exterior equivalence*, *converse*, *isometric isomorphism*, *conjugate actions*, and *coefficient automorphism*. The source and nearby primary arXiv papers on twisted or transformation-group L^p crossed products were inspected. No paper explicitly resolving the literal question or presenting this counterexample was found.

The novelty confidence is moderate: the example is elementary and the conjugacy obstruction is standard in spirit. More importantly, this packet does not settle a strengthened rigidity problem in which the isomorphism must fix the coefficient algebra, preserve its canonical copy, or preserve a grading/coaction. A reviewer should first confirm that the unqualified phrase “an isometric isomorphism” in the source is meant literally. The main mathematical audit point is the pairing of coefficient representations in (3).

References

- [1] A. Delfin, C. Farsi, and J. Packer, *Twisted crossed products of Banach algebras*, arXiv:2509.24106, 2025 (version inspected: May 2026), <https://arxiv.org/abs/2509.24106>.